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A Portable Myoelectric Robotic System for Light Exercise Among Bedridden and Wheelchair Bound Individuals

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Abstract— In this study an interactive exercise technique is developed to cater to bedridden and wheelchair bound individuals, in order for them to remain both mentally and physically alert despite their immobility. The sedentary lifestyle common among bedridden and wheelchair bound individuals leads to a variety of problems such as stiff muscles and numbness in various parts of the body. To perform this light exercise, surface electromyogram (EMG) signals are measured from major muscles from the body responsible for human motion. These measured signals are processed and further used to control the movements of a mobile robot (either left, right or straight) wirelessly using radio frequency transmission.

Keywords— Electromyogram, Biceps Brachii, amplification, filtering and RF Transmission.

I. INTRODUCTION

In this paper we propose using a mobile robot that responds to muscle activity as a means for exercise to incorporate some degree of physical activity among bedridden and wheelchair bound individuals. In tennis for example, the player is active physically by swinging the racket to hit the ball. At the same time the player is also mentally active because he has to see the ball and decide exactly when and where to hit the ball with his racket. In this proposed robotic based exercise method, the patient exercises physically by navigating the mobile robot's movements using his left and right bicep muscle activity. At the same time the patient is also mentally active because he or she needs to observe the mobile robot's movements (left, right or forward) and decide which direction it should move to achieve the target point. Surface Electromyogram (EMG) signals measured from the left and right bicep serve as the muscle activity indicator of the bedridden or wheelchair bound individual. EMG signal refers to the electrical current generated upon muscle contraction [1]. When muscle fibers contract, there is an exchange of ions within the muscle fiber membranes, this flow of ions generates an electrical current [1] [2]. Flow of ions within muscle fiber membranes can be measured on the surface of the skin by using Argentum Chloride surface electrodes which convert ion flow into electron flow.

II. SYSTEM DESCRIPTION

Fig 1 shows the two main components for this system namely the transmitter block and receiver block. At the transmitter block, the EMG signals from the individual are obtained, processed, digitized and transmitted via radio frequency. At the receiver block, the digitized EMG signal transmitted by the transmitter is received by a radio frequency receiver. This received signal is used to control the movements of the mobile robot forward, left or right in real time.

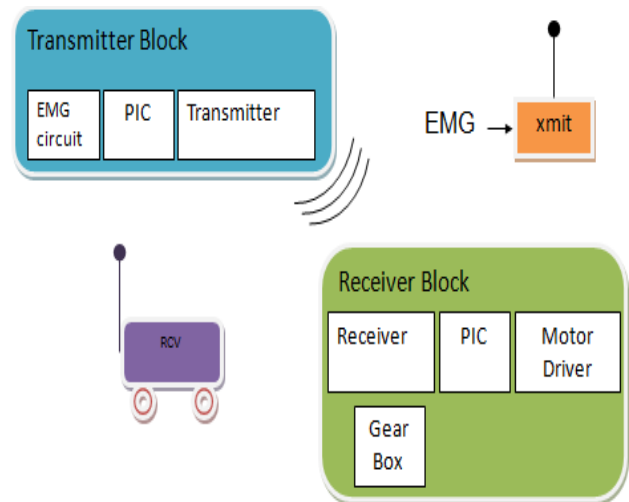


Fig. 1 Transmitter and Receiver blocks for this system

III. METHODS

A. EMG Signal Conditioning

EMG signals are measured from the skin surface of the left and right biceps brachii muscles as shown in Fig 3. Bipolar electrode configuration is used where two surface EMG electrodes are placed on the muscle of interest and another electrode is placed on a bony part of the body, such as the elbow [3][4].

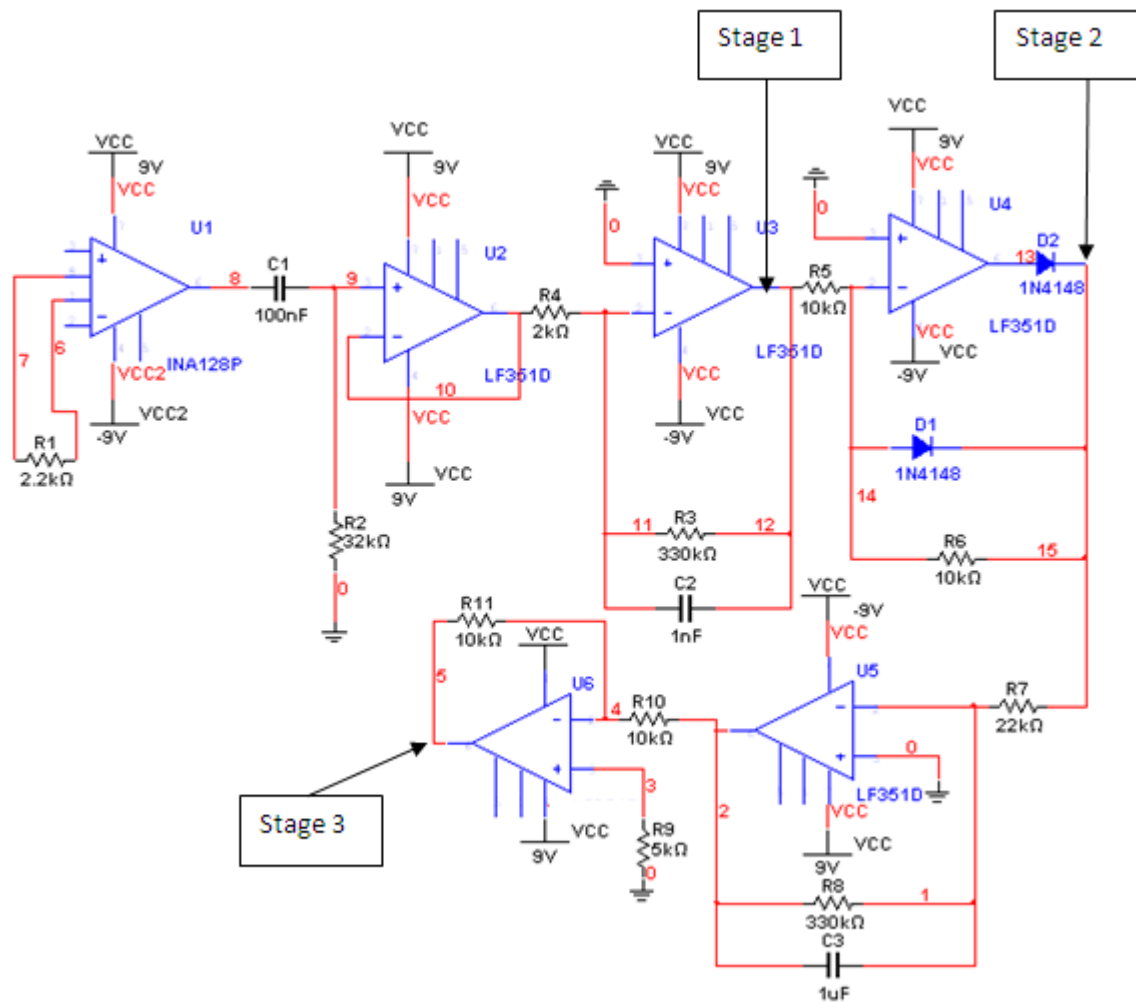


Fig. 2 EMG Signal Conditioning Schematic

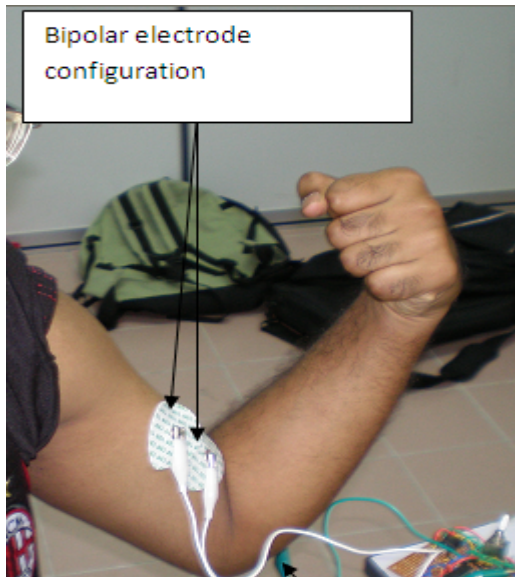


Fig. 3 Electrodes placed on the left biceps brachii muscle

The placement of the electrode on a bony part of the body serves as a good grounding point, since there are no muscles fiber bundles on the elbow [4]. The EMG signal conditioning consists of an instrumentation amplifier, band pass filter, precision half wave rectifier and envelope detector [5]. The circuit schematic is shown in Fig 2. Stage 1 is the output of the band pass filter, stage 2 is the output of the half wave rectifier and stage 3 inverted smoothing circuit output.

1) Instrumentation Amplifier:

In this paper we used the INA 128P which has a high CMRR value of 120dB [6]. The high CMRR is necessary to suppress noise from the surroundings such as the 50Hz power line interference which is a common mode signal to both the instrumentation amplifier inputs. Another useful feature of the INA 128P is that the gain can conveniently set by connecting an external resistor between pins 1 and 8 of the INA 128 [6].

2) Band Pass Filtering (50-500Hz) and Rectification:

The output of the instrumentation amplifier is fed to an active band pass filter which consists of a high pass filter with 50Hz cutoff frequency and a low pass filter with 500Hz cut off frequency [7]. Both the low pass and high pass filters were constructed using either LM324 or LF351 op amps. The 50Hz high pass filter suppresses frequencies due to motion artifact whereas the low pass filter with 500Hz cut off frequency suppresses frequencies due to signal aliasing [8]. The resulting signal after band pass filtering (Fig 2, stage1) is shown in Fig 8. The EMG signal conditioning output will be sent to a microcontroller which only accepts voltages within the 0-5V TTL range. Therefore the output of the band pass filter is rectified using a precision half wave rectifier circuit (Fig 2, stage2) [9]. As a result the negative portion of the EMG signal is eliminated as shown in Fig. 9.

3) Smoothing/Envelope Detection:

The final stage in EMG signal conditioning is smoothing (Fig 2, stage 3). The output waveform for the smoothing stage is shown in Fig 10. Smoothing is done in order for the EMG signal to resemble muscle force exerted by the individual [10]. This step is crucial for our application where the robot's movements are dependent on the patients muscle activity. Smoothing is accomplished by passing the output of the precision half wave rectifier in the previous stage through an active low pass filter with very low cut off frequency [10]. After experimentation, we found that a cut off frequency of 0.48Hz for the smoothing low pass filter gives the smoothest signal from our subject's left and right biceps brachii muscles. This cut off frequency for the low pass filter may vary between individuals.

B. Wireless Transmission of EMG Signal and Robot Control

The smoothed EMG signal from the left and right biceps are fed to the analog inputs pins of a PIC16F877A microcontroller [11]. The RF wireless transmitter is also connected to this PIC16F877A, consequently the analog EMG signals are digitized by the PIC16F877A and expressed in numerical values that range from 0 (0V) to 255 (5V). These numerical values are then sent to the Xbee wireless transceiver which serves as a transmitter [12]. The digitized EMG signals are transmitted wirelessly using Radio Frequency transmission at 2.4 GHz. Xbee RF wireless modules are used to accomplish simplex transmission of the EMG signals. There is another Xbee transceiver on the receiver PIC16F877A, a program is written in the PIC-C compiler to accept the wirelessly transmitted EMG signals from the Xbee receiver. This program drives the motors (via the L293D motor driver) when the EMG signal has reached a threshold value.

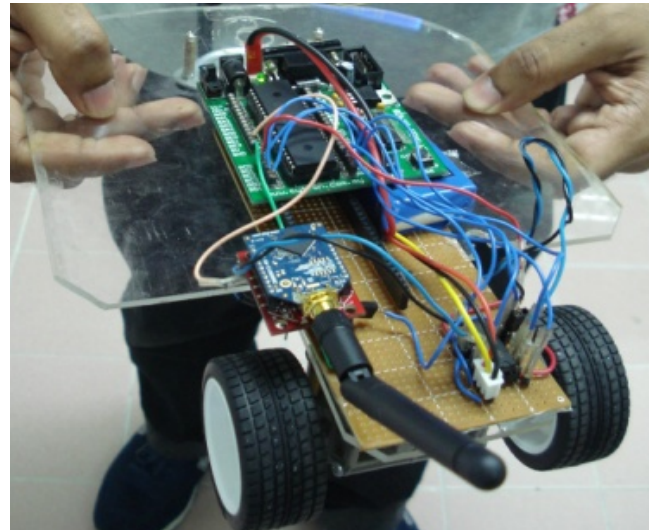


Fig. 4 A simple mobile robot constructed using a tamiya gear box

The robot moves in three modes. The first mode occurs when the left bicep is contracted. Due to the left biceps brachii muscle contraction, the amplitude of the smoothed EMG signal exceeds a specified threshold. The program written in the PIC16F877A microcontroller instructs the right motor to move and the left motor to stop, causing the robot to turn towards the left direction as shown in Fig 5. Similarly due to the right biceps brachii muscle contraction, the amplitude of the smoothed EMG signal exceeds a specified threshold. The program written in the PIC16F877A microcontroller instructs the left motor to move and the right motor to stop, causing the robot to turn towards the right direction as shown in Fig 6. The third mode occurs when the amplitude of the EMG signal from both the left and right biceps brachii muscles have exceeded the threshold; as a result both the left and right motors move causing the robot to move straight as shown in Fig 7.



Fig. 5 The robot moves towards the left when the left biceps brachii muscle is contracted.



Fig. 6 The robot moves towards the right when the right biceps brachii muscle is contracted.



Fig. 7 The robot moves straight when both left and right biceps brachii muscles are contracted

IV. RESULTS AND DISCUSSION

A. Signal Conditioning Output

1) Band-pass Filter:

After band-pass filtering as shown in Fig 8, we can see that when a muscle is flexed, the amplitude of the EMG signal increases due to the increase in flow of ions within the muscle fiber membranes. From this signal we can differentiate when the muscle is flexed and when the muscle is at rest. However the PIC16F877A microcontroller can only accept voltage levels within the 0-5V TTL range, therefore further processing is required to remove the negative frequency content.

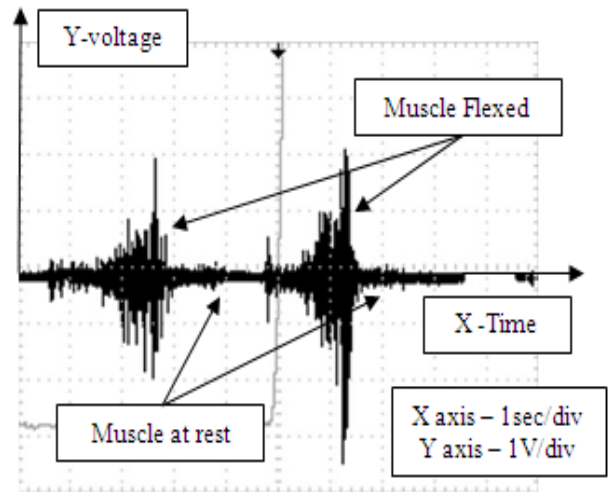


Fig. 8 Band-pass filtered EMG signal (oscilloscope capture)

2) Half-wave rectification:

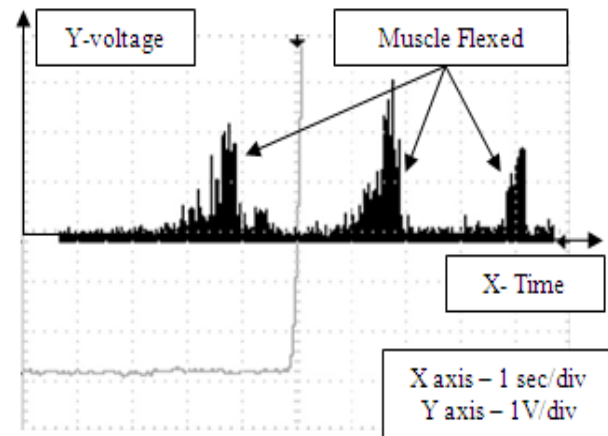


Fig. 9 Half wave rectified EMG signal (oscilloscope capture)

Half wave rectification as shown in Fig 9 does eliminate frequency components of the EMG signal. If we are interested in maintaining the frequency components of the EMG signal, full wave rectification is necessary [9]. Full wave rectification keeps the negative portion of the signal by reflecting the negative part of the signal onto the positive part [9].

3) Smoothing/Envelope Detection:

The smoothing stage as shown in Fig 10 completes the EMG signal processing aspect in this paper. Similar to half wave rectification, smoothing is also a process which removes frequency content of the EMG signal. There is a tradeoff between signal smoothness (determined by low pass filter cut off frequency) and response time of the EMG signal. The smoother the signal, the slower its response time [10].

Passing the signal through the smoothing circuit we had used attenuated the signal. To compensate for signal attenuation, the smoothing circuit also has its own gain. The two resistors on the smoothing low pass filter provide amplification with gain = $330/22 = 16.5$. The smoothing low pass filter circuit inverts the EMG signal due to its negative gain. An op amp inverter converts the negative signal into a positive signal.

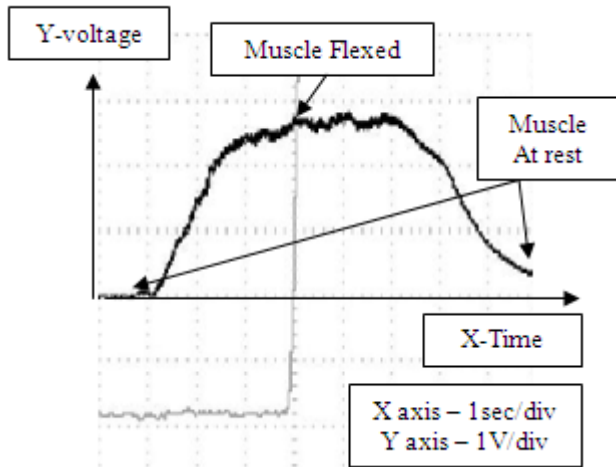


Fig. 10 Smoothing circuit output (oscilloscope capture).

B. Factors Effecting EMG Signal Quality

1) EMG Signal Clipping:

If the gain of the instrumentation amplifier, band pass filter or the combined gain of the instrumentation amplifier and band pass filter is too high, the EMG signal will be clipped due to op-amp saturation [13]. Fig 8 shows a rectified EMG signal which has been clipped. Clipped EMG signals no longer give a good indication of the individuals muscle activity, in Fig 11 voltage beyond 2V is not visible even though our subject was contracting his bicep with more effort.

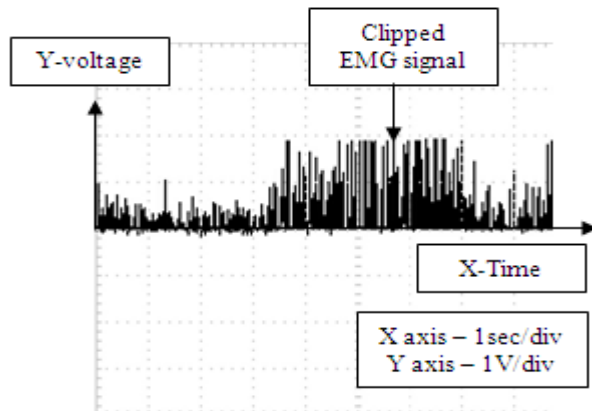


Fig. 11 Clipped EMG Signal (oscilloscope capture)

To prevent the EMG signal from being clipped off, the signal is amplified in multiple stages. In this paper the gain of the instrumentation amplifier was 30 whereas the gain of the band pass filter was 100, resulting in a total gain of 3000 (excluding the gain of the smoothing circuit required to compensate for signal attenuation).

2) EMG Signal Offset:

In this paper, the EMG signal amplification, filtering and smoothing made the EMG signal suitable to be inputted to the next stage which is the analog input on the PIC16F877A microcontroller. However after passing through the smoothing circuit, the EMG signal we obtained may be subject to offset as shown in Fig 12.

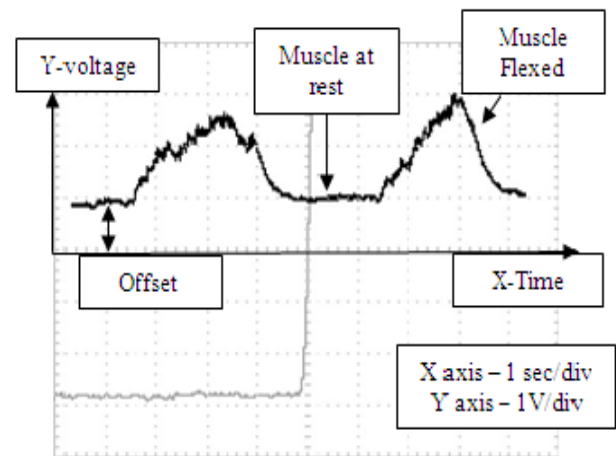


Fig. 12. An offset in the EMG signal after smoothing (oscilloscope capture).

When the muscle is at rest, the amplitude of the signal should be near to zero. However from Fig 12 we can see that the value when muscle is at rest is 1V. To solve this offset problem, we can use an op amp bias adjustment circuit.

3) EMG Signal Contaminated by noise:

While measuring the EMG signals, our readings were contaminated by noise as shown in Fig 13 below.

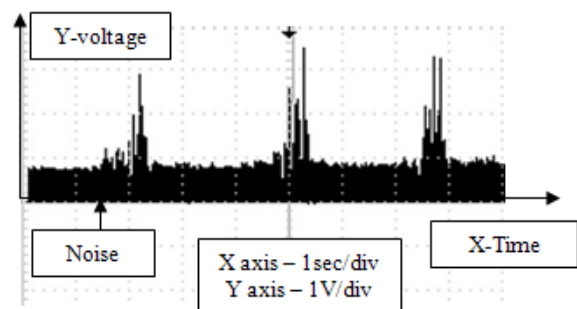


Fig. 13 Rectified EMG signal contaminated by noise

The reason for this noise contamination was due to the distance between the two wires connecting the electrodes to the inputs pins (pin 2 and 3) of the INA 128 instrumentation amplifier. To suppress this noise we twisted the two wires together, keeping these two wires close to each other and soldered them directly to pins 2 and 3 of the INA 128P. Keeping these two wires close to each other helps the instrumentation amplifier suppress common mode signals.

V. CONCLUSION

An EMG signal based exercise method for bedridden and wheelchair bound individuals is proposed. The paper explores the possibility of controlling a mobile robot wirelessly using EMG signals as one of the means in the efforts to make bedridden and wheelchair bound individuals both physically and mentally active.

VI. FUTURE WORK

In order to verify this systems applicability, we need to test the system on bedridden and wheelchair bound individuals. In this initial stage we had only tested the system on healthy individuals.

From the smoothed EMG signal or the EMG quasi-tension [14], we can get a good indication of how much effort was exerted by the muscles using the integrated EMG technique [2]. Physicians can use data from the integrated EMG as an indicator of how physically active the patient was throughout the exercise routine. Finding the EMG quasi-tension is another method to make the EMG signal resemble muscle force [14].

In this study, we targeted the biceps brachii muscles, however in order for patients to exercise other major muscle groups involved in human motion should be considered such as tibialis anterior, gastrocnemius, adductor longus, extensor carpi radialis and flexor carpi radialis. Due to the random and stochastic properties of the EMG signal, it is common for researchers to normalize the EMG signal by finding the Maximum Voluntary Contraction (MVC). EMG signals are represented as percentage of MVC.

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